

Inferring Starlink Latency Structure from Public RIPE Atlas Measurements

Jiaxian Tu¹, Yuxuan Li¹, Shenghe Xu^{2,†}, Lifan Mei^{1,*}

¹School of Advanced Technology, Xi'an Jiaotong-Liverpool University ²Amazon.com, Inc.

{Jiaxian.Tu23, Yuxuan.Li2302}@student.xjtlu.edu.cn

shenghexu@gmail.com, Lifan.Mei@xjtlu.edu.cn

Abstract—Most Starlink measurement studies rely on owned dishes, browser extensions, or active scans of the terrestrial network—all of which require infrastructure that an outside reader cannot easily reproduce. We ask the opposite question: how much can be inferred about Starlink latency from public RIPE Atlas built-in pings alone, and where does this approach fall short? Our data combines 897,949 ping samples from 122 RIPE Atlas probes on SpaceX AS14593 (Starlink’s autonomous system) over a 7-day window, the public Starlink TLE catalogue, and a 42-entry corpus of publicly disclosed Starlink-related entities (FCC-licensed earth stations, peering/PoP locations inferred from public BGP, known IXP sites). We find three things. First, per-probe median RTT (round-trip time) groups into four broad latency clusters with centers near 23, 42, 71, and 125 ms. Second, the great-circle distance to the nearest disclosed entity, d_{pub} , is moderately correlated with RTT (Spearman $\rho_s = 0.40$). The slowest cluster contains both probes that are geometrically far from any disclosed entity (Marshall Is., Guam) and probes whose RTT is not explained by d_{pub} alone (Mongolia, Iran, Jordan). Third, built-in pings fire around every 240 s. This sampling rate is too sparse to resolve the 15 s satellite handover events reported by owned-dish studies. Two limitations apply throughout: root-DNS targets are anycast, so RTT mixes Starlink’s path with DNS routing; and the public infrastructure corpus is sparse outside North America and Europe.

I. INTRODUCTION

Starlink is now the largest commercial LEO constellation (>7,000 active satellites [1]). Starlink performance studies have used owned dishes and browser extensions [2], active scanning of the terrestrial network [3], or re-analyses of third-party measurement platforms [4]. Prior public-data work has already noted regional latency variation and the effect of ground-station proximity.

We focus narrowly on what RIPE Atlas built-in pings can show about Starlink. RIPE Atlas is a global measurement network operated by RIPE NCC; built-in pings to root DNS servers run continuously from every probe and require no new measurement setup on our side [5]. Our contribution is a concrete map of what RIPE Atlas does recover (coarse latency groups, a moderate distance–RTT relationship) and the specific cases where it does not (slow probes that public infrastructure cannot explain, handover events at sub-second timescales).

II. DATASETS AND METHOD

Probe corpus. On 2026-05-20 the RIPE Atlas API [5] reports 125 active IPv4 probes on AS14593 (Starlink’s

autonomous system); 122 had $\geq 2,400$ successful samples over our 7-day window. Each probe pings three IPv4 root DNS anchors (k.root, f.root, b.root; msm-ids 1001/1004/1010) approximately every 240 s. We collect 897,949 successful samples across 39 countries.

Anycast caveat. The three root-DNS targets are *anycast*, so RTT mixes Starlink’s access path with the DNS routing decision at sample time. Replication against fixed non-anycast endpoints (e.g., specific RIPE Atlas anchors) is a natural follow-up.

Starlink TLEs. 2026-05-20 CelesTrak snapshot [1] contains 10,360 catalogued elements; we propagate with SGP4 when needed.

Public infrastructure corpus. We compile 42 publicly disclosed Starlink-related entities of three kinds: (a) licensed earth-station gateways from FCC IBFS/ICFS filings [6]; (b) peering points and PoPs (Points of Presence) for AS14593 inferred from public BGP (Border Gateway Protocol) data—indicative of network attachment, not authoritative ground-station locations; (c) known IXP (Internet Exchange Point) sites. Coverage is dense in NA/EU and sparse elsewhere. For each probe we compute d_{pub} , the great-circle distance to the nearest of these 42 entities.

Group discovery. The main metric is per-probe median RTT across the three anchors. We run 1-D k -means on $\log_{10}(\text{median RTT})$. The short/medium/long ordering is qualitatively stable across $k \in \{3, 4, 5\}$; we use $k=4$ as the working choice since it isolates the longest-latency cases. At $k=4$ the cluster centers are 23.4, 42.1, 71.2, 125.4 ms and data-driven boundaries are $\{31, 55, 95\}$ ms; we label the four groups **T1–T4** in order of increasing RTT.

III. FINDINGS

F1: Four broad latency groups. Fig. 2 (left) shows the per-group median-RTT distribution; k -means centers ($\star$) sit within 5 ms of the box medians, and group sizes are $\{77, 21, 19, 5\}$. The same ordering appears at $k=3$ and $k=5$; we report $k=4$ because it isolates the smallest, most informative group, not because four is the only valid choice.

F2: d_{pub} correlates moderately with RTT. Fig. 2 (middle): Spearman $\rho_s = 0.40$ ($p < 10^{-5}$). A threshold $d_{\text{pub}} \geq 2,500$ km selects 7 probes, 6 in T3/T4 (point precision $\sim 86\%$; the $n=7$ exact 95% binomial confidence interval is $\approx [42\%, 100\%]$). We use this as a *directional* signal: distance to public infrastructure tracks RTT, but not tightly enough to act as a precise

*Corresponding author. [†]Work done outside of Amazon.

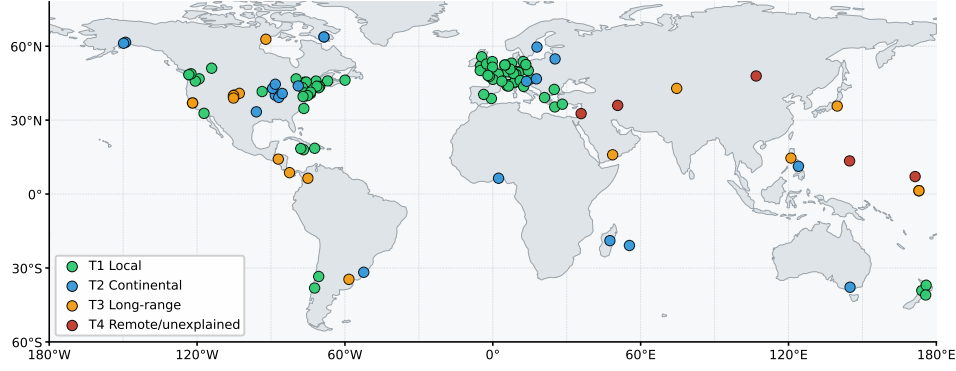


Fig. 1. 122 public RIPE Atlas probes on AS14593, colored by inferred latency group. T4 (red) splits into two cases: probes geometrically far from any of our 42 disclosed entities (Marshall Is., Guam) and probes reachable to a disclosed entity yet not explained by d_{pub} alone (Mongolia, Iran, Jordan). See Table I.

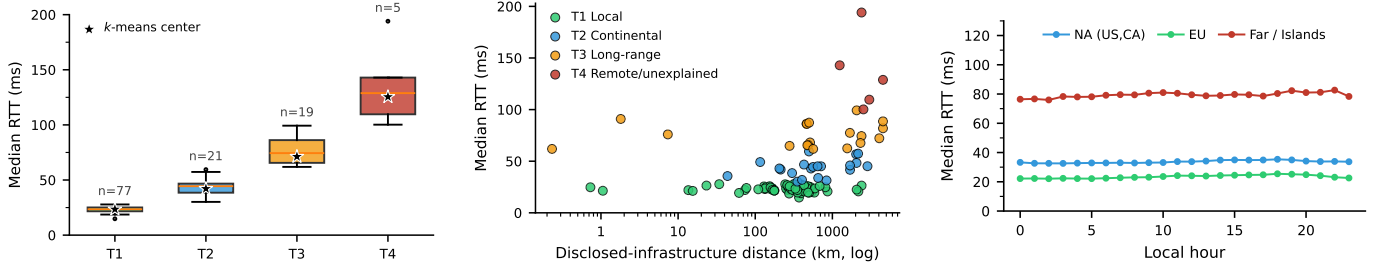


Fig. 2. **Left:** per-group median-RTT distribution with k -means centers ($\star$). **Middle:** median RTT vs. d_{pub} (great-circle distance to the nearest of 42 publicly disclosed Starlink-related entities, log scale). Spearman $\rho_s = 0.40$ ($p < 10^{-5}$). T3/T4 concentrate in the upper-right; several T4 probes (JO, IR, MN) sit at moderate d_{pub} and are not explained by distance alone. **Right:** median RTT by longitude-corrected local hour for three regional aggregates—EU $\sim 20\%$ evening rise, NA $\sim 10\%$, remote-island/Far probes diurnally flat.

TABLE I

THE FIVE T4 PROBES, SEPARATED BY MECHANISM. CC = ISO 3166-1 ALPHA-2 COUNTRY CODE (MH = MARSHALL ISLANDS, GU = GUAM, MN = MONGOLIA, IR = IRAN, JO = JORDAN). “GEOMETRIC” = NO DISCLOSED ENTITY WITHIN PLAUSIBLE BENT-PIPE RANGE; “UNEXPLAINED” = ENTITY REACHABLE, YET OBSERVED RTT EXCEEDS WHAT d_{pub} ALONE PREDICTS.

prb id	cc	d_{pub} (km)	RTT (ms)	Mechanism
64237	MH	4,528	128.8	geometric
65337	GU	2,525	100.2	geometric
1010672	MN	3,010	109.6	unexplained
1012845	IR	2,389	194.1	unexplained
1002312	JO	1,244	142.9	unexplained

classifier. The 2,500 km cut is the observed inflection on this dataset; the exact value is sensitive to which entities enter the corpus.

Table I distinguishes two cases. Marshall Islands and Guam fall outside plausible reach of any disclosed entity; the public corpus is consistent with the observed RTT. Mongolia, Iran, and Jordan are reachable to a disclosed entity yet observed RTT exceeds what d_{pub} alone predicts. These are *not explained by public infrastructure distance*; the likely explanations are unobserved routing/peering constraints or missing infrastructure visibility—not direct evidence of a policy cause.

F3: Built-in pings cannot see 15 s handovers. RIPE Atlas built-in pings fire approximately every 240 s with sub-2 s jitter. The 15 s satellite handover known from owned dish studies [2] requires sub-second cadence aligned to the handover boundary, and pooled across-probe samples ($P(\Delta_t > 20 \text{ ms}) \sim 1.5\%$) do not break through residual noise. This is an undersampling

issue, not a fundamental sampling-theoretic limit, but it is decisive in practice: higher-cadence user-defined measurements ($\leq 1 \text{ s}$) are required.

IV. LIMITATIONS AND VALIDATION PATH

Limits. (L1) Anycast root-DNS targets mean reported RTT mixes Starlink’s access path with DNS routing. (L2) The 42-entity corpus combines gateways, PoPs, peering points, and IXPs of varying authority, and is sparse outside North America and Europe; d_{pub} is therefore a proxy for distance to *public* infrastructure, not to the actual Starlink path. (L3) The $d_{pub} \geq 2,500 \text{ km}$ flag rests on 7 probes—directional, not a calibrated classifier. (L4) Seven-day medians smooth diurnal and orbital variation. **Next:** non-anycast fixed-location anchors (L1); broaden the public infrastructure corpus, e.g. via community-maintained gateway lists (L2); bootstrap CIs and longer windows (L3, L4); $\leq 1 \text{ s}$ user-defined measurements for handover detection (F3).

REFERENCES

- [1] T.S. Kelso, “CelesTrak: Starlink Group TLE Catalogue,” <https://celestrak.org/NORAD/elements/gp.php?GROUP=starlink>, 2026, accessed 2026-05-20.
- [2] N. Mohan, A. E. Ferguson, H. Cech *et al.*, “A Multifaceted Look at Starlink Performance,” in *Proc. ACM Web Conf. (WWW)*, 2024.
- [3] Y. Wu, M. Cui, G. Gou *et al.*, “Beneath the Heavens: A Thorough Measurement Study of the Starlink Terrestrial Network,” in *Proc. IEEE/ACM IWQoS*, 2025.
- [4] R. A. U. Richter, V. Ververis, and V. Bajpai, “Breaking Through the Clouds: Performance Insights into Starlink’s Latency and Packet Loss,” in *Proc. IFIP Networking*, 2025.
- [5] RIPE NCC, “RIPE Atlas: Built-in Measurements,” <https://atlas.ripe.net/docs/getting-started/built-in-measurements>, 2026.
- [6] U.S. FCC, “FCC IBFS/ICFS Earth Station Filings (Starlink Services LLC),” <https://www.fcc.gov/icfs/>, 2026.